

# SEBASTIÁN HERMOSILLA CANOBRA

## Bachelor of Science in Astronomy

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## STUDIES

### Bachelor of Science in Astronomy

#### University of Chile

 March 2019 – July 2023    Santiago, Chile

### School Study

#### German Institute of Puerto Montt

 March 2004 – Dec. 2018    Puerto Montt, Chile

## RESEARCH EXPERIENCE

### Asteroids Research Project

#### Department of Astronomy, U. Chile

 August 2023-Present    Santiago, Chile

- This research focuses on detecting asteroids within the Earth's orbit. The methodology involves conducting observational campaigns during twilight periods. Tutor: Prof. Patricio Rojo.

### Planetary Atmospheres Research Project

#### Department of Astronomy, U. Chile

 August 2022-August 2023    Santiago, Chile

- The aim is to create a catalog with the atmospheric properties of various solar system bodies to compare them with future observations of exoplanets. Tutor: Prof. Patricio Rojo.

### Distant Galaxy Research Project

#### Department of Astronomy, U. Chile

 Jan. - Dec. 2022    Santiago, Chile

- From a numerical simulation we seek to predict the age of galaxies at  $z \sim 4$ , under the context of the James Webb Space Telescope (JWST). Tutor: Prof. Valentino González.

## OTHER EXPERIENCES

### Teaching Assistant

#### Department of Physics, U. Chile

 March-July 2023    Santiago, Chile

- Introduction to Classical Physics: Responsible for grading tests and conducting laboratory classes. Teacher: Valentino González.

### Observational Experience

#### Association of Universities for Research in Astronomy (AURA)

 July 2022    Cerro Tololo Inter-American Observatory

- Conducting observations of planetary transits using the 0.9-meter SMARTS Telescope over the course of a week.

## CAREER GOALS

*"Build knowledge through observation, research and teaching."*

## ACHIEVEMENTS



### FCFM Outstanding Student (3x)

Selected as an outstanding student of the Faculty of Physical and Mathematical Sciences of University of Chile in 2020, 2021 and 2022.



### National PSU score

2018 - Obtained the highest score in the mathematics test for university admission, along with the highest score in Los Lagos region (Chile) in the science test.

### Math Olympiad Champion

2018 - First place in Mathematics Olympiads organized by the San Sebastián University.



### University Cross Country and Athletics Team

2023 - Third place in the National Championship for Universities as a member of the Cross Country team.

### National Champion

2015 - National Champion as a member of the Puerto Montt city basketball team.



### Picture of the Month (2x)

I have been honored twice with the "Analog Picture of the Month" award from Migo Laboratory in 2021 and 2022.

## STRENGTHS

Creativity

Proactivity

Teamwork

Responsibility

Solidarity

## SKILLS

Data analysis & visualization

Numerical problem solving

Solid physical and mathematical basis

Machine learning

## COURSES

Workshop - Science and Commit

Ph.D. Candidates in Astronomy at the University of Chile

📅 March 2022

📍 remote

- Knowledge was acquired to learn how to work collaboratively using the GitHub platform.

Basic Mountaineering Course

University Andean Club

📅 August 2020 - Dec. 2020

📍 Andes Mountains, Chile

- I have gained essential skills for mountain activities, encompassing displacement techniques, orientation to navigation, first aid and more.

## OTHER INTERESTS

Apart from science, I have a range of other interests:



### University Track and Field Team

Specialized in 10,000m and 5,000m races, during 2022 and 2023.



### Engineering and Sciences Diffusion Group

Led efforts to bridge university research with Chilean society and promote faculty enrollment among school students. Engaged since 2022.



### TECHO for Chile Foundation Member

Volunteer in the construction of housing for communities in crisis in 2019 and 2020.



### Analog Photography

Since 2019 I have developed a passion for analog photography.



### Basketball

Chosen to represent Puerto Montt city's team and actively played for the city's club from 2015 to 2017. Throughout my academic journey, I consistently contributed to my school's basketball team, as well.

## LANGUAGES

Spanish (Native)

English (Advanced)

German (Basic)



## REFERENCES

Prof. Patricio Rojo

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🏢 Department of Astronomy, U Chile

Prof. Valentino González

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🏢 Department of Astronomy, U. Chile